

Normalization for the Voting Management System Database

Introduction

Normalization is the process of organizing data in a database to reduce redundancy, avoid anomalies, and improve data integrity. In this project, the database is normalized up to Boyce-Codd Normal Form (BCNF).

The normalization process is explained step-by-step from Unnormalized Form (UNF) to BCNF.

Problems in UNF

- Data redundancy exists.
- Candidate, voter, party, and region information are repeated many times.
- Update anomalies may occur.
- Insert and delete anomalies may occur.
- Difficult to maintain consistency.

Therefore, normalization is required.

2. First Normal Form (1NF)

A relation is in First Normal Form when:

- All attributes contain atomic values.
- There are no repeating groups.
- Each record is unique.

1NF Relations

REGION

region_code (PK)	region_name	
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VOTER

voter_id (PK)	full_name	date_of_birth	gender	address
national_id	phone_number	email	status	registered_at

region_code (FK)				
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POLLING_STATION

polling_station_id (PK)	station_name	station_code	address	latitude	longitude
region_code (FK)					

PARTY

party_id (PK)	party_name
party_abbreviation	registration_number
symbol_url	description

CANDIDATE

candidate_id (PK)	candidate_name	date_of_birth	gender	address
photo_url	status	party_id (FK)		

ADMIN

admin_id (PK)	full_name	email	username	password_hash	role	last_login
created_at						

VOTE

vote_id (PK)	vote_date
is_valid	voter_id (FK)
party_id (FK)	polling_station_id (FK)

At this stage:

- All attributes are atomic.
- Repeating groups are removed.
- Each table has a primary key.

However, partial dependency and transitive dependency may still exist.

3. Second Normal Form (2NF)

A relation is in Second Normal Form when:

- It is already in 1NF.
- No partial dependency exists.
- Non-key attributes depend on the whole primary key.

In this project, most tables already use single-attribute primary keys. Therefore, partial dependency is automatically removed.

Functional Dependencies

REGION

region_code → region_name

VOTER

voter_id → full_name, date_of_birth, gender, address, national_id, phone_number, email, status, registered_at, region_code

national_id → voter_id

email → voter_id

phone_number → voter_id

POLLING_STATION

polling_station_id → station_name, station_code, address, latitude, longitude, region_code

station_code → polling_station_id

PARTY

party_id → party_name, party_abbreviation, registration_number, symbol_url, description

party_abbreviation → party_id

registration_number → party_id

CANDIDATE

candidate_id → candidate_name, date_of_birth, gender, address, photo_url, status, party_id

ADMIN

admin_id → full_name, email, username, password_hash, role, last_login, created_at

email → admin_id

username → admin_id

VOTE

vote_id → vote_date, is_valid, voter_id, party_id, polling_station_id

Since all non-key attributes depend fully on the primary key, the schema is in 2NF.

4. Third Normal Form (3NF)

A relation is in Third Normal Form when:

- It is already in 2NF.
- No transitive dependency exists.
- Non-key attributes do not depend on other non-key attributes.

Checking Transitive Dependencies

REGION

No transitive dependency exists.

VOTER

All attributes depend directly on voter_id.

No transitive dependency exists.

POLLING_STATION

All attributes depend directly on polling_station_id.

No transitive dependency exists.

PARTY

All attributes depend directly on party_id.

No transitive dependency exists.

CANDIDATE

All attributes depend directly on candidate_id.

No transitive dependency exists.

ADMIN

All attributes depend directly on admin_id.
No transitive dependency exists.

VOTE

All attributes depend directly on vote_id.
No transitive dependency exists.

Therefore, the database is in 3NF.

5. Boyce-Codd Normal Form (BCNF)

A relation is in BCNF when:

- It is already in 3NF.
- Every determinant is a candidate key.

BCNF Verification

REGION

Determinant:

- region_code → region_name

region_code is a candidate key.

Therefore, REGION is in BCNF.

VOTER

Determinants:

- voter_id → all voter attributes
- national_id → voter_id
- email → voter_id
- phone_number → voter_id

All determinants are candidate keys because national_id, email, and phone_number are unique.
Therefore, VOTER is in BCNF.

POLLING_STATION

Determinants:

- $\text{polling_station_id} \rightarrow \text{all attributes}$
- $\text{station_code} \rightarrow \text{polling_station_id}$

Both are candidate keys.

Therefore, POLLING_STATION is in BCNF.

PARTY

Determinants:

- $\text{party_id} \rightarrow \text{all attributes}$
- $\text{party_abbreviation} \rightarrow \text{party_id}$
- $\text{registration_number} \rightarrow \text{party_id}$

All determinants are candidate keys.

Therefore, PARTY is in BCNF.

CANDIDATE

Determinant:

- $\text{candidate_id} \rightarrow \text{all attributes}$

candidate_id is a candidate key.

Therefore, CANDIDATE is in BCNF.

ADMIN

Determinants:

- $\text{admin_id} \rightarrow \text{all attributes}$
- $\text{email} \rightarrow \text{admin_id}$
- $\text{username} \rightarrow \text{admin_id}$

All determinants are candidate keys.

Therefore, ADMIN is in BCNF.

VOTE

Determinant:

- $\text{vote_id} \rightarrow \text{vote_date, is_valid, voter_id, party_id, polling_station_id}$

vote_id is a candidate key.
Therefore, VOTE is in BCNF.

Final BCNF Relations

REGION

- region_code (PK)
- region_name

VOTER

- | | | |
|-----------------|-------------------------|--------------------|
| • voter_id (PK) | • address | • email (UNIQUE) |
| • full_name | • national_id (UNIQUE) | • status |
| • date_of_birth | | • registered_at |
| • gender | • phone_number (UNIQUE) | • region_code (FK) |

POLLING_STATION

- | | | |
|---------------------------|-------------------------|--------------------|
| • polling_station_id (PK) | • station_code (UNIQUE) | • latitude |
| • station_name | • address | • longitude |
| | | • region_code (FK) |

PARTY

- | | |
|-------------------------------|--------------------------------|
| • party_id (PK) | • registration_number (UNIQUE) |
| • party_name | • symbol_url |
| • party_abbreviation (UNIQUE) | • description |

CANDIDATE

- | | | |
|---------------------|-------------|-----------------|
| • candidate_id (PK) | • gender | • status |
| • candidate_name | • address | • party_id (FK) |
| • date_of_birth | • photo_url | |

ADMIN

- | | |
|-----------------|-------------|
| • admin_id (PK) | • full_name |
|-----------------|-------------|

- email (UNIQUE)
- username (UNIQUE)
- password_hash
- role
- last_login
- created_at

VOTE

- vote_id (PK)
- vote_date
- is_valid
- voter_id (FK)
- party_id (FK)
- polling_station_id (FK)

Conclusion

The Voting Management System database was normalized from UNF up to BCNF. During the normalization process:

- Redundant data was removed.
- Data consistency was improved.
- Insert, update, and delete anomalies were minimized.
- Data integrity and maintainability were enhanced.

The final database structure is efficient, scalable, and suitable for implementation in both MySQL and MongoDB systems.